



$$1.) m(a + bX) = a + b \times m(X)$$

$$m(a + bX) = \frac{1}{N} \sum_{i=1}^N (a + b x_i)$$

$$\begin{array}{c} \frac{a}{N} + b \cdot \frac{1}{N} \sum_{i=1}^N x_i \\ \text{const} \downarrow \quad \underbrace{\hspace{10em}} \\ = a + b \times m(X) \quad \checkmark \end{array}$$

$$2.) \text{cov}(X, a+bY) = b \times \text{cov}(X, Y)$$

$$\text{cov}(X, a+bY) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))(a+bY - m(a+bY))$$

Ignore a const.

$$\frac{1}{N} \sum_{i=1}^N (x_i - m(X)) b (Y - m(Y))$$

$$b \cdot \frac{1}{N} \sum_{i=1}^N (x_i - m(X)) (Y - m(Y))$$

$$= b \times \text{cov}(X, Y)$$

$$3) \text{cov}(a+bX, a+bX) = b^2 \text{cov}(X, X)$$

$$= \text{cov}(a+bX, a+bX) = \frac{1}{N} \sum_{i=1}^N (a+bX - m(a+bX))^2$$

↓

$$= \frac{1}{N} \sum_{i=1}^N b^2 (X - m(X))^2$$

$$= b^2 \frac{1}{N} \sum_{i=1}^N (X - m(X))^2$$

↓

s^2

$$s^2 = \text{cov}(X, X)$$

$$= b^2 \text{cov}(X, X) \quad \checkmark$$

4.) No it is not always true that $m(g(x)) = g(m(x))$ because the output from the value of x will be differentiated based on which function it is being plugged into first.

